

what features are supported in a browser. In this particular case it turns out that from IE9 onwards the “addEventListener” method is in fact supported in the DOM. If you choose to do browser detection as we have done in the snippet above then it basically means that the onus of tracking what features are supported in what browser and in what version is upon you! A far better alternative is to use feature detection. The same functionality given above can be achieved using feature detection like so:

```
• if (window.addEventListener) {
•     window.addEventListener("load", onLoad);
• } else if(window.attachEvent) {
•     window.attachEvent("onload", onLoad);
• }
```

In this case it really does not matter what browser or version the user is using, if “addEventListener” is available then it will be used. One library that really helps implement feature detection in an efficient manner is Modernizr which simplifies the process of testing for availability of various features. Testing whether canvas is supported for instance looks like this:

```
• if (Modernizr.canvas) {
•     // yep, canvas supported!
• }
```

Internet Explorer 9 compatibility modes

Internet Explorer 9 displays webpages that contain the HTML5 document type in IE9 Standards mode, which provides the highest support available for established and emerging industry standards, including the HTML5 (Working Draft), W3C Cascading Style Sheets Level 3 Specification (Working Draft), Scalable Vector Graphics (SVG) 1.0 Specification, and others.

In some cases, existing websites may not display correctly when displayed in IE9 mode. This can happen for any number of reasons, including (but not limited to) the following:

- ▶ The design of a site may rely on the behavior of a specific version of a browser and that behavior has changed in later versions of the browser, such as conditional comments, version vectors, and user-agent detection.
- ▶ The design of a site may rely on non-standard behavior.
- ▶ The design of a site may rely on functionality that is no longer supported in the latest versions of various standards or browsers.

The design of a site may rely on functionality that was incorrectly implemented in an earlier version of a given browser.